Exponentials & Polynomials

Section 1: Laws of Exponents & Radicals

1 If $x > 0$, which expression is equivalent to $\sqrt[3]{x^6 \cdot x^9}$?

a. x^5

b. x^3

c. x^{15}

d. x^6

2 If $2^{3x-1} = 32$, what is the value of x?

a. 1

b. 2

c. 3

d. 4

3 Which of the following is equivalent to $\frac{x^{5/2}}{x^{1/2}}$ for all $x > 0$?

a. x^2

b. x^3

c. x^5

d. $x^{5/4}$

Section 2: Exponential Growth and Decay Models

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4 A town had a population of $12,000$ in the year 2010 and increases by 3.5% each year. Which function models the population $P(t)$, where t is the number of years since 2010?

a. $P(t) = 12000(1.35)^t$

b. $P(t) = 12000(1.035)^t$

c. $P(t) = 12000(0.965)^t$

d. $P(t) = 12000 + 35t$

5 A certain medication decays in the bloodstream according to $A(t) = 500(0.75)^t$, where $A(t)$ is in milligrams and t is in hours. By what percentage does the amount of medication decrease each hour?

a. 75%

b. 25%

c. 50%

d. 15%

6 An investment account triples in value every 8 years. If the initial balance is 1,500 dollars, which expression represents the account balance after t years?

a. $1500(3)^{t/8}$

b. $1500(3)^{8t}$

c. $1500(8)^{t/3}$

d. $1500(1.03)^{8t}$

Section 3: Polynomial Operations & Factoring

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7 Which expression is the quotient when $2x^3 - 5x^2 + 7x - 4$ is divided by $x - 1$?

a. $2x^2 - 3x + 4$

b. $2x^2 + 3x - 4$

c. $2x^2 - 7x + 4$

d. $2x^2 - 3x - 4$

8 If polynomial $P(x) = x^3 - kx^2 + 4x - 12$ has a zero at $x = 3$, what is the value of k?

a. 2

b. 3

c. 4

d. 5

9 Which of the following is equivalent to $(3x - 2)(x^2 + 4x - 5)$?

a. $3x^3 + 10x^2 - 23x + 10$

b. $3x^3 + 12x^2 - 15x + 10$

c. $3x^3 - 2x^2 + 8x - 10$

d. $3x^3 + 10x^2 - 15x - 10$

Section 4: Polynomial Graphs & Roots

10 A polynomial function $f(x)$ has zeros at $x = -2, 1, 4$. Which equation could define f?

a. $f(x) = (x + 2)(x - 1)(x - 4)$

b. $f(x) = (x - 2)(x + 1)(x + 4)$

c. $f(x) = (x + 2)^2(x - 1)$

d. $f(x) = x^3 - 2x + 4$

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11 The graph of $y = (x - 3)^2(x + 1)$ touches the x-axis without crossing it at which value of x?

a. $x = -1$

b. $x = 3$

c. $x = 0$

d. $x = 1$

12 What is the degree of the polynomial function $g(x) = (x^2 - 1)(2x - 3)^3$?

a. 4

b. 5

c. 6

d. 7

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